

MICHAEL MULUGETA DEMISSIE

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SUMMARY

AI/ML engineer who builds and ships production agentic AI systems — voice agents, RAG pipelines, and production ML models — backed by 4+ years of deep backend engineering experience. Currently evaluating and improving frontier LLMs on technical tasks (rubric-based scoring, training-data authoring, agentic evaluation) for AI training programs, while building and freelancing on AI-powered products end to end. Strong attention to detail, fast iteration, and clear technical writing.

CORE SKILLS

- AI / Agents & LLMs** Agentic AI, LLM APIs (Groq, OpenAI, HuggingFace), RAG, Vector Databases (pgvector), BERT/DistilBERT, Prompt Engineering, Agent/Voice Pipelines (Twilio, Deepgram, ElevenLabs)
- AI Training & Evaluation** LLM Evaluation, Rubric-Based Scoring, Response Ranking, Training-Data Authoring, Agentic Task Evaluation, Structured Feedback
- Backend & Systems** Python, Node.js, FastAPI, Express, REST APIs, Microservices, System Design, Async/WebSockets
- Data & Infra** PostgreSQL, MySQL, Redis, Apache Airflow, Docker, AWS (S3, Lambda), CI/CD, Railway, Render
- ML & Languages** PyTorch, scikit-learn, LightGBM, Pandas, NumPy, Java, JavaScript, SQL, Bash

EXPERIENCE

- AI Trainer — Software Engineering Expert** | **Handshake AI Fellowship** May 2026 – Present
 - Develop and evaluate domain-specific prompts to assess frontier LLM performance on software-engineering tasks; rank outputs for correctness, efficiency, and clarity using rubric-based scoring.
 - Author training data — code solutions and structured instructions — that guide model behavior on technical tasks, and evaluate agentic AI on multi-step, real-world computer-use tasks.
- Expert Software Developer** | **micro1** May 2026 – Present
 - Execute and test production LLMs on software-engineering tasks across multiple rounds, documenting failures and submitting evaluated outputs with structured written feedback against detailed technical guidelines.
- Software Engineer** | **Hasab AI** Jun 2024 – Jun 2025

FastAPI · PostgreSQL · Redis · Docker · AWS

 - Built and maintained the API layer for a voice-intelligence platform delivering speech-to-text, translation, and text-to-speech for low-resource African languages, designing endpoints to support transcription pipelines.
- Software Engineer Intern** | **ParallelDots** Jun 2024 – Nov 2024

Node.js · Express · PostgreSQL · MySQL

 - Built and debugged client-facing REST API endpoints for an AI computer-vision platform serving Fortune 500 clients; integrated third-party services supporting image-processing features.
- Freelance Software Engineer** | **Self-Employed** Feb 2022 – Present

Python · FastAPI/Flask · Node.js · React · PostgreSQL/MySQL

 - Delivered 10+ full-stack projects and 100+ assistive engagements for startups and individual clients — backend APIs, web apps, and automation scripts, managing scope and delivery independently.

SELECTED PROJECTS

- Voice Bot QA Agent — Real-Time AI Phone Agent Tester** | [github](#) Jun 2026

Twilio · Deepgram · Groq · ElevenLabs · WebSockets

 - Built a real-time voice agent that places live phone calls and role-plays 10 patient personas to stress-test a healthcare scheduling AI; diagnosed and fixed live audio, turn-taking, and self-silencing bugs; produced a systemic QA report from automated and human-reviewed call analysis.
- DocuQuery — Production RAG Pipeline & AI Document Assistant** | [github](#) · [live](#) Apr – Jun 2026

FastAPI · pgvector · HuggingFace · Groq · Airflow · Docker

 - Built a full-stack RAG app with streaming, document-grounded answers; engineered chunking, embeddings, and an HNSW-indexed pgvector store, with a daily Airflow ingestion DAG from a public jobs API.
- FinResolve — AI Complaint Routing & Priority Triage** | [github](#) · [live](#) Jan – May 2026

DistilBERT · LightGBM · FastAPI · PostgreSQL · AWS S3

 - Built and deployed a production ML pipeline routing 1M+ CFPB complaints to the correct department (DistilBERT) and a priority tier (TF-IDF + LightGBM, 88% accuracy) behind a JWT-authenticated FastAPI service with role-based views.

EDUCATION

M.S. Data Analytics, George Washington University (Expected 2027) | **B.Tech Software Engineering**, Delhi Technological University (2025)